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Learning Implicit Credit Assignment for Cooperative Multi-Agent Reinforcement Learning

Meng Zhou* **Ziyu Liu*** **Pengwei Sui** **Yixuan Li** **Yuk Ying Chung**
The University of Sydney

Abstract

We present a multi-agent actor-critic method that aims to implicitly address the credit assignment problem under fully cooperative settings. Our key motivation is that credit assignment among agents may not require an explicit formulation as long as (1) the policy gradients derived from a centralized critic carry sufficient information for the decentralized agents to maximize their joint action value through optimal cooperation and (2) a sustained level of exploration is enforced throughout training. Under the centralized training with decentralized execution (CTDE) paradigm, we achieve the former by formulating the centralized critic as a hypernetwork such that a latent state representation is integrated into the policy gradients through its multiplicative association with the stochastic policies; to achieve the latter, we derive a simple technique called *adaptive entropy regularization* where magnitudes of the entropy gradients are dynamically rescaled based on the current policy stochasticity to encourage consistent levels of exploration. Our algorithm, referred to as LICA, is evaluated on several benchmarks including the multi-agent particle environments and a set of challenging StarCraft II micromanagement tasks, and we show that LICA significantly outperforms previous methods.

1 Introduction

Many complex real-world problems such as autonomous vehicle coordination [3], network routing [55], and robot swarm control [17] can naturally be formulated as multi-agent cooperative games, where reinforcement learning (RL) presents a powerful and general framework for training robust agents. Though single-agent RL algorithms can be trivially applied to these environments by treating the multi-agent system as a single actor with a joint action space, the limited scalability and the inherent constraints on agent observability and communication in many multi-agent environments necessitate *decentralized* policies that act only on their local observations. A straightforward approach to learn decentralized policies is to train the agents independently, but the simultaneous exploration of the agents often results in non-stationary environments where learning becomes highly unstable. To this end, previous work relies on a standard paradigm known as *centralized training with decentralized execution* (CTDE) [34, 25, 37, 10, 28, 4], where the independent agents can access additional state information that is unavailable during policy inference.

However, one major challenge of CTDE in cooperative settings is *credit assignment*, which refers to the task of attributing a global, shared reward from the environment to the agents' individual actions. Solving the credit assignment problem explicitly may give useful insights into which agents or agent actions were responsible for the collective reward signal and may thus substantially facilitate policy optimization, but doing so is often nontrivial since the interactions between the agents and the environment can be highly complex. A notable approach is to assess individual agent actions by calculating *difference rewards* against a certain reward baseline [48, 36, 10], but these methods can

*Equal contribution. Author ordering is random. Correspondence to Meng Zhou <mzho7212@gmail.com> and Ziyu Liu <kenziyuliu@outlook.com>.

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be inefficient as they require separate estimations for the baselines and can become less effective with complex cooperation behaviors. Another line of approach is to represent the global state-action value as a (rule-based or learned) aggregation of the individual state-action values [46, 37, 45, 28]. A representative method is QMIX [37], which achieves *implicit* credit assignment by learning a non-linear *mixing network* that conditions on the global state and maps the individual agent Q-values into the joint action Q-value estimate. While these methods allow more complex credit assignment, the capacity of the value mixing network is still limited by the monotonic relationships between the joint and the individual Q-values. Their extensions to continuous action spaces may also require additional strategies that can compromise performance and/or complexity.

In this work, we propose a policy-based algorithm called LICA for learning implicit credit assignment that aims to address the above limitations. LICA is closely related to the family of *value gradient* methods [7, 52, 15, 43, 14, 26, 2] where policies are directly optimized in the direction of the approximate state/action value gradients. Apart from applying the framework to multi-agent settings, our key contribution is to extend the concept of value mixing [46, 37, 28, 45] to *policy mixing*, where the centralized critic is formulated as a hypernetwork [11] that maps the current state information into a set of weights which, in turn, mixes the individual action vectors into the joint action value estimate. Compared to previous policy-based methods such as [10, 27, 18], this practical formulation introduces an extra latent state representation into the policy gradients to provide sufficient information for learning optimal cooperative behaviors without explicit credit assignment strategies. It also trivially achieves higher expressiveness than value mixing methods as there are no inherent constraints on the mixing weights of the centralized critic. Following the above motivation, we also explore an alternative training regime where continuous approximations of action samples from the stochastic policies are replaced with explicit action distribution parameters to provide more information about the agents' behaviors during policy optimization. While the resulting action value gradients can be less accurate, we observe that policies can learn stably and often converge faster to better solutions.

One notable challenge for policy-based algorithms is maintaining consistent levels of exploration to prevent premature convergence to sub-optimal policies [53, 30, 1]. Many existing methods [42, 13, 18, 27, 49] address this with entropy regularization, where the policy entropy is added to the training objective to favor more stochastic actions. While widely adopted, we argue that the vanilla form of entropy regularization could be ineffective for this purpose due to the undesirable curvature of the entropy function derivative. To this end, we further propose a simple technique called *adaptive entropy regularization* where the magnitudes of entropy gradients are inversely adjusted based on the policy entropy itself. We show that this allows easier tuning of the regularization strength and more consistent levels of policy stochasticity throughout training.

We benchmark our methods on two sets of cooperative environments, the Multi-Agent Particle Environments [27] and the StarCraft Multi-Agent Challenge [39], and we observe considerable performance improvements over previous state-of-the-art algorithms. We also conduct further component studies to demonstrate that (1) compared to difference reward based credit assignment approaches (e.g. [10]), LICA has higher representational capacity and can readily handle environments where multiple global optima exist, and (2) our adaptive entropy regularization is crucial for encouraging sustained exploration and can lead to faster policy convergence in complex scenarios.

2 Related Work

Explicit Credit Assignment. In general, explicit methods provide strategies for attributing agent contributions that are at least provably locally optimal [24]. COMA [10] is a representative method that uses a centralized critic to estimate the counterfactual advantage of an agent action, but it quickly becomes ineffective with complex cooperation behaviors. SQDDPG [49] provides a theoretical framework where credit assignment is based on an agent's approximate *marginal contribution* as it is sequentially added to the agent group. While theoretically justified, this formulation assumes an initial oracle scheduling the agents with global observation that is often unavailable in practice.

Implicit Credit Assignment. In contrast, implicit methods do not purposely assess the individual actions against a certain baseline, and most previous methods address credit assignment by directly learning a value decomposition from the shared reward signal into the individual component value functions [46, 37, 45, 28]. An earlier work is VDN [46], where the value decomposition is linear and the state information is ignored during training. QMIX [37] presents an improvement by conditioning

a hypernetwork on the global state for a non-linear mixing of the individual action-values, but it is still limited by the monotonicity constraint of its mixing weights. QTRAN [45] attempts to address these limitations with a provably more general value factorization, but it imposes computationally intractable constraints that, if relaxed, can lead to poor empirical performance [28]. In contrast, the policy-based LICA is practical and has high capacity as no additive or monotonic constraints exist.

Model-Free Value Gradients and Deterministic Policy Gradients. As our method learns the decentralized policies by backpropagating the joint action value, it is related to the family of value gradient [52, 7, 8, 15, 14] and deterministic policy gradient (DPG) [43, 26, 27, 49] algorithms. On a high level, both classes of methods aim to improve the policy by directly ascending the gradients of (approximate) state-values (V) or action-values (Q) via chain rule [14]. In particular, DPG methods assume deterministic policies to efficiently estimate policy gradients, and they can be considered the deterministic limit of the model-free variant of stochastic value gradient (i.e. SVG(0)) [15], a type of value gradient methods that deploys *reparameterization* (e.g. [23, 38, 20]) to similarly allow backpropagation of action value gradients. Among existing methods, MADDPG [27] extends the DPG framework to multi-agent settings, where each agent learns a deterministic policy from joint action value gradients and maintains a separate (one for every agent) centralized MLP critic network for learning different reward structures. In contrast, our method learns stochastic policies from the centralized critic (one for all agents) formulated as a hypernetwork, which is crucial for better utilization of the state for learning complex agent cooperation without credit assignment strategies.

Entropy Regularization is a common technique for improving policy optimization by inducing more stochastic actions and possibly a smoother objective landscape [1], where many existing methods [53, 12, 30, 40, 42, 13, 18, 27, 49] augment the training objective with a weighted entropy loss term. Others also consider, e.g., decaying schedules for the regularization strength [32, 1]. As mentioned in Section 1, we argue that its common form could be ineffective at keeping consistent exploration levels that are often required for uncovering optimal joint actions in challenging environments.

3 Methods

3.1 Preliminaries and Notations

In this work we primarily focus on learning policies in discrete action spaces, though extensions to continuous action spaces are possible. We consider a fully cooperative multi-agent task with n agents $\mathcal{A} = \{1, ..., n\}$ as a Dec-POMDP [33] defined by a tuple $G = (S, U, P, r, Z, O, n, \gamma)$. At each time step t, each agent a chooses an action u_t^a from its action space $U_a \in \{U_1, ..., U_n\} \equiv U$, forming a joint action $u_t \in (U_1 \times \cdots \times U_n) \equiv U^n$. $P(s_{t+1}|s_t, u_t)\colon S \times U^n \times S \to [0, 1]$ is the state transition function where S is the set of true states and $s_t \in S$ is the state at time t. $r(s_t, u_t)\colon S \times U^n \to \mathbb{R}$ is the reward function yielding a shared reward r_t for u_t, and $\gamma \in [0, 1)$ is the discount factor. We consider partially observable settings, so each agent a acts on its local observation $z_t^a \in Z$ drawn from the observation function $O(s_t, a)\colon S \times \mathcal{A} \to Z$ with observation space Z. To select an action, each agent a follow its stochastic policy $\pi^a(u_t^a|z_t^a)\colon Z \times U_a \to [0, 1]$, and the agents' joint policy π induces a joint action value function $Q^\pi(s_t, u_t) = \mathbb{E}_\pi[R_t|s_t, u_t]$ where $R_t = \sum_{t'=t}^{T} \gamma^{t'-t} r_{t'}$ is the discounted accumulated reward with finite horizon T. Our goal is to find the optimal joint policy π^* such that $Q^{\pi^*}(s_t, u_t) \geq Q^\pi(s_t, u_t)$ for all π and $(s_t, u_t) \in S \times U^n$. We also use $Q^\pi(s_t, u_t)$ and $Q^\pi(s_t, u_t^1, ..., u_t^n)$ interchangeably and use $\tau = (s_1, z_1, u_1, r_1, s_2, ...)$ to denote a trajectory. Function approximations of π and Q^π are parameterized by $\theta = \{\theta_1, ..., \theta_n\}$ and ϕ respectively.

3.2 LICA: Learning Implicit Credit Assignment for Cooperative Environments

In this section, we present a new method called LICA that aims to implicitly address multi-agent credit assignment under shared rewards in fully cooperative environments. Our key motivation is that if the decentralized policies can be directly optimized to maximize the true joint action value function, then any converged policies should have acquired (at least locally) optimal cooperative behaviors without *explicitly* formulating a credit assignment among agents, which can be unrealistic and unscalable in complex scenarios. In practice, if a trained critic $Q_\phi^\pi(s, u)$ provides a close approximation of Q^π for the current π_θ, then finding an end-to-end differentiable optimization setting where the policies simultaneously improve along the joint action value gradients $\nabla_\theta Q_\phi^\pi(s, u)$ should act as a proxy for finding optimal credit assignment strategies (e.g., studied in COMA [10]).